

Full Title of the Research Paper

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Abstract—This paper presents a complete IEEE two-column paper template with working code snippets for figures, flowcharts, tables, multi-line equations, and matrix arrays.

Index Terms—Flowcharts, LaTeX Templates, IEEEtran, Tables, Mathematical Equations.

I. INTRODUCTION

This template contains examples of images, flowcharts using TikZ, tables, matrix arrays, and system equations. Historical foundational work in computing was documented by Moore [?], while modern neural architecture concepts build upon Vaswani et al. [?] and Goodfellow et al. [?].

II. IMAGES AND FIGURES

Standard single-column image insertion using `graphicx`:

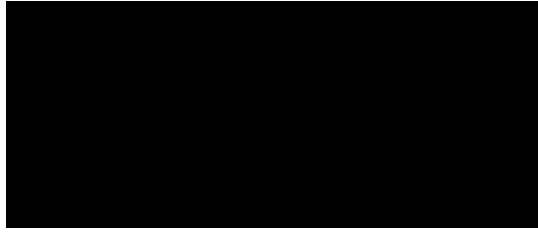


Fig. 1. Example of a single-column image.

III. FLOWCHARTS

Flowchart generated directly in LaTeX using the `tikz` package:

IV. EQUATIONS AND ARRAYS

An aligned multi-line equation:

$$f(x) = ax^2 + bx + c \quad (1)$$

$$g(x) = \int_0^\infty e^{-x^2} dx \quad (2)$$

A matrix array using `amsmath`:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad (3)$$

V. TABLES

An IEEE-formatted table with merged rows and columns:

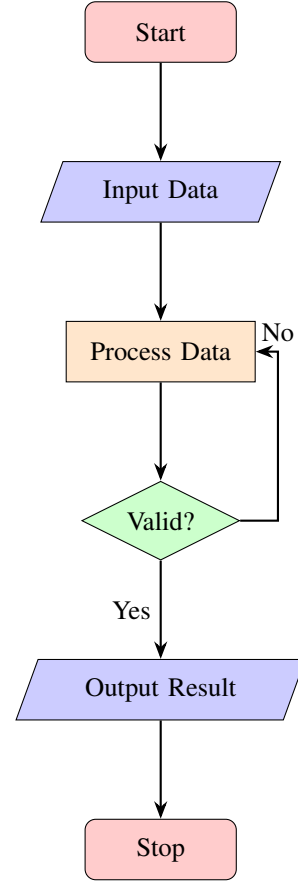


Fig. 2. Process pipeline flowchart.

TABLE I
PERFORMANCE COMPARISON OF SYSTEM MODELS

Model	Accuracy (%)	Precision	Recall	F1-Score
Model A	92.4	0.91	0.93	0.92
Model B	95.1	0.94	0.95	0.94
Proposed	97.8	0.97	0.98	0.97
	98.3	0.98	0.99	0.98

VI. CONCLUSION

This setup provides a complete workspace for drafting an IEEE paper directly inside Overleaf without missing dependencies.

REFERENCES